Assignment NO 5:

SQL Queries on: Functional Single Row, Aggregate functions, Data Sorting, Subqueries, Group by having, Set Operations, View, TCL Commands.

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Class - SYCSE

Panel – C [Batch C3]

Problem Statement:

Travel Agency Booking System

A travel agency, "Voyage Ventures," wants to analyze its booking and customer data to identify trends and optimize travel packages. The database consists of the following tables:

- **customers: customer_id, first_name, last_name, email, city**
- **bookings: booking_id, customer_id, package_id, booking_date, total_price**
- **packages: package_id, package_name, destination, duration_days**
- **employees: employee_id, first_name, last_name, hire_date, department**

mysql> create database voyage_ventures;

Query OK, 1 row affected (0.03 sec)

```
mysql> use voyage_ventures;
Database changed
mysql> create table customers (
    -> customer_id int primary key,
    ->  first_name varchar(50) not null,
    -> last_name varchar(50) not null,
    -> email varchar(100),
    -> city varchar(50)
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql> create table packages (
    ->     package_id int primary key,
    ->     package_name varchar(100),
    ->     destination varchar(100),
    ->     duration_days int
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> create table employees (
    ->     employee_id int primary key,
    ->     first_name varchar(50),
    ->     last_name varchar(50),
    ->     hire_date date,
    ->     department varchar(50)
    -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> create table bookings (
    ->     booking_id int primary key,
    ->     customer_id int,
    ->     package_id int,
    ->     employee_id int,
    ->     booking_date date,
    ->     total_price decimal(10,2),
    ->     foreign key (customer_id) references customers(customer_id),
    ->     foreign key (package_id) references packages(package_id),
    ->     foreign key (employee_id) references employees(employee_id)
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql> desc customers;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| customer_id | int          | NO   | PRI | NULL    |       |
| first_name  | varchar(50)  | NO   |     | NULL    |       |
| last_name   | varchar(50)  | NO   |     | NULL    |       |
| email       | varchar(100) | YES  |     | NULL    |       |
| city        | varchar(50)  | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
5 rows in set (0.02 sec)
```

```
mysql> desc packages;
+---------------+--------------+------+-----+---------+-------+
| Field         | Type         | Null | Key | Default | Extra |
+---------------+--------------+------+-----+---------+-------+
| package_id    | int          | NO   | PRI | NULL    |       |
| package_name  | varchar(100) | YES  |     | NULL    |       |
| destination   | varchar(100) | YES  |     | NULL    |       |
| duration_days | int          | YES  |     | NULL    |       |
+---------------+--------------+------+-----+---------+-------+
4 rows in set (0.00 sec)

mysql> desc employees;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| employee_id | int         | NO   | PRI | NULL    |       |
| first_name  | varchar(50) | YES  |     | NULL    |       |
| last_name   | varchar(50) | YES  |     | NULL    |       |
| hire_date   | date        | YES  |     | NULL    |       |
| department  | varchar(50) | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
5 rows in set (0.01 sec)
```

```
mysql> desc bookings;
+--------------+---------------+------+-----+---------+-------+
| Field        | Type          | Null | Key | Default | Extra |
+--------------+---------------+------+-----+---------+-------+
| booking_id   | int           | NO   | PRI | NULL    |       |
| customer_id  | int           | YES  | MUL | NULL    |       |
| package_id   | int           | YES  | MUL | NULL    |       |
| employee_id  | int           | YES  | MUL | NULL    |       |
| booking_date | date          | YES  |     | NULL    |       |
| total_price  | decimal(10,2) | YES  |     | NULL    |       |
+--------------+---------------+------+-----+---------+-------+
6 rows in set (0.00 sec)

mysql> insert into customers values
    -> (101, 'Amit', 'Sharma', 'amit@gmail.com', 'Pune'),
    -> (102, 'Sneha', 'Patil', 'sneha@gmail.com', 'Mumbai'),
    -> (103, 'Rahul', 'Verma', 'rahul@gmail.com', 'Delhi'),
    -> (104, 'Priya', 'Mehta', 'priya@gmail.com', 'Chennai'),
    -> (105, 'Rohan', 'Kulkarni', 'rohan@gmail.com', 'Bangalore');
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> insert into packages values
    -> (201, 'Europe Tour', 'Paris', 12),
    -> (202, 'Dubai Trip', 'Dubai', 7),
    -> (203, 'Thailand Escape', 'Bangkok', 10),
    -> (204, 'Maldives Luxury', 'Maldives', 5),
```

```
    -> (205, 'Singapore Special', 'Singapore', 8);
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> insert into employees values
    -> (301, 'Neha', 'Joshi', '2023-01-15', 'Sales'),
    -> (302, 'Arjun', 'Reddy', '2022-03-10', 'Marketing'),
    -> (303, 'Kiran', 'Desai', '2021-07-20', 'Sales'),
    -> (304, 'Meera', 'Kapoor', '2024-02-01', 'Support');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> insert into bookings values
    -> (701, 101, 201, 301, curdate() - interval 2 month, 60000),
    -> (702, 102, 202, 302, curdate() - interval 1 month, 45000),
    -> (703, 103, 201, 301, curdate() - interval 4 month, 70000),
    -> (704, 104, 203, 303, curdate() - interval 7 month, 30000),
    -> (705, 105, 205, 304, curdate() - interval 3 month, 55000);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from customers;
+-------------+------------+-----------+-----------------+-----------+
| customer_id | first_name | last_name | email           | city      |
+-------------+------------+-----------+-----------------+-----------+
|         101 | Amit       | Sharma    | amit@gmail.com  | Pune      |
|         102 | Sneha      | Patil     | sneha@gmail.com | Mumbai    |
```

```
|         103 | Rahul      | Verma      | rahul@gmail.com | Delhi      |
|         104 | Priya      | Mehta      | priya@gmail.com | Chennai    |
|         105 | Rohan      | Kulkarni   | rohan@gmail.com | Bangalore  |
+-------------+------------+-----------+-----------------+-----------+
5 rows in set (0.00 sec)

mysql> select * from packages;
+------------+-------------------+-------------+---------------+
| package_id | package_name      | destination | duration_days |
+------------+-------------------+-------------+---------------+
|        201 | Europe Tour       | Paris       |            12 |
|        202 | Dubai Trip        | Dubai       |             7 |
|        203 | Thailand Escape   | Bangkok     |            10 |
|        204 | Maldives Luxury   | Maldives    |             5 |
|        205 | Singapore Special | Singapore   |             8 |
+------------+-------------------+-------------+---------------+
5 rows in set (0.00 sec)
```

```
mysql> select * from employees;

+-------------+------------+-----------+------------+------------+

| employee_id | first_name | last_name | hire_date  | department |

+-------------+------------+-----------+------------+------------+

|         301 | Neha       | Joshi     | 2023-01-15 | Sales      |

|         302 | Arjun      | Reddy     | 2022-03-10 | Marketing  |

|         303 | Kiran      | Desai     | 2021-07-20 | Sales      |

|         304 | Meera      | Kapoor    | 2024-02-01 | Support    |

+-------------+------------+-----------+------------+------------+

4 rows in set (0.00 sec)


mysql> select * from bookings;

+------------+-------------+------------+-------------+--------------+-------------+

| booking_id | customer_id | package_id | employee_id | booking_date | total_price |

+------------+-------------+------------+-------------+--------------+-------------+

|        701 |         101 |        201 |         301 | 2026-01-03   |    60000.00 |

|        702 |         102 |        202 |         302 | 2026-02-03   |    45000.00 |

|        703 |         103 |        201 |         301 | 2025-11-03   |    70000.00 |

|        704 |         104 |        203 |         303 | 2025-08-03   |    30000.00 |

|        705 |         105 |        205 |         304 | 2025-12-03   |    55000.00 |

+------------+-------------+------------+-------------+--------------+-------------+

5 rows in set (0.00 sec)
```

1. Customer & Package Analysis (Functional Single Row & Data Sorting)

The marketing team needs a report of all bookings made in the last 6 months. The report should display for each booking:

- **The customer's full name in lowercase.**
- **The destination with a "Destination: " prefix (e.g., "Destination: Paris").**
- **A calculated estimated_return_date based on the booking date and package duration. The report should be sorted by the booking date in descending order.**

```
mysql> select
    ->     lower(concat(c.first_name, ' ', c.last_name)) as customer_full_name,
    ->     concat('Destination: ', p.destination) as destination,
    ->     date_add(b.booking_date, interval p.duration_days day) as
estimated_return_date
    -> from bookings b
    -> join customers c on b.customer_id = c.customer_id
    -> join packages p on b.package_id = p.package_id
    -> where b.booking_date >= date_sub(curdate(), interval 6 month)
    -> order by b.booking_date desc;
+--------------------+------------------------+-----------------------+
| customer_full_name | destination            | estimated_return_date |
+--------------------+------------------------+-----------------------+
| sneha patil        | Destination: Dubai     | 2026-02-10            |
| amit sharma        | Destination: Paris     | 2026-01-15            |
| rohan kulkarni     | Destination: Singapore | 2025-12-11            |
| rahul verma        | Destination: Paris     | 2025-11-15            |
```

```
+--------------------+-----------------------+----------------------+
4 rows in set (0.01 sec)
```

2. Sales & Revenue (Aggregate Functions, GROUP BY, HAVING)

The finance team needs to analyze revenue from different destinations. They need a summary report that calculates the total revenue, average booking price, and the total number of bookings for each destination. The report should only include destinations that have generated a total revenue of more than $50,000.

```
mysql> select
    ->     p.destination,
    ->     sum(b.total_price) as total_revenue,
    ->     avg(b.total_price) as average_booking_price,
    ->     count(b.booking_id) as total_bookings
    -> from bookings b
    -> join packages p on b.package_id = p.package_id
    -> group by p.destination
    -> having sum(b.total_price) > 50000;
+-------------+---------------+-----------------------+----------------+
| destination | total_revenue | average_booking_price | total_bookings |
+-------------+---------------+-----------------------+----------------+
| Paris       |     130000.00 |          65000.000000 |              2 |
| Singapore   |      55000.00 |          55000.000000 |              1 |
+-------------+---------------+-----------------------+----------------+
2 rows in set (0.01 sec)
```

3. Employee & Customer Link (Subqueries)

The HR department wants to recognize employees who are associated with the most valuable customers. A valuable customer is one whose total booking spend is above the average total booking spend of all customers. The task is to list the full names and departments of all employees who have handled bookings for these valuable customers.

```
mysql> select distinct
    ->     e.first_name,
    ->     e.last_name,
    ->     e.department
    -> from employees e
    -> join bookings b on e.employee_id = b.employee_id
    -> where b.customer_id in (
    ->     select customer_id
    ->     from bookings
    ->     group by customer_id
    ->     having sum(total_price) > (
    ->         select avg(total_spend)
    ->         from (
    ->             select sum(total_price) as total_spend
    ->             from bookings
    ->             group by customer_id
    ->         ) as temp
    ->     )
    -> );
```

```
+------------+-----------+------------+
| first_name | last_name | department |
+------------+-----------+------------+
| Neha       | Joshi     | Sales      |
| Meera      | Kapoor    | Support    |
+------------+-----------+------------+
2 rows in set (0.01 sec)
```

4. Data Consolidation (Set Operations)

The operations team needs a master list of individuals for a customer satisfaction survey. They need a single list of names (first and last names) that combines:

- **All customers who have booked a package with a duration of more than 10 days.**
- **All employees who have been hired in the last two years. The list should not contain any duplicate names.**

```
mysql> select first_name, last_name
    -> from customers
    -> where customer_id in (
    ->     select b.customer_id
    ->     from bookings b
    ->     join packages p on b.package_id = p.package_id
    ->     where p.duration_days > 10
    -> )
    ->
    -> union
    ->
```

```
    -> select first_name, last_name
    -> from employees
    -> where hire_date >= date_sub(curdate(), interval 2 year);
+------------+-----------+
| first_name | last_name |
+------------+-----------+
| Amit       | Sharma    |
| Rahul      | Verma     |
+------------+-----------+
2 rows in set (0.00 sec)
```

5. Simplified Access (View)

To simplify daily reporting, the sales manager wants to create a virtual table. This VIEW, named v_customer_bookings_summary, should show the booking details for each customer. It should include the customer's full name, the package name they booked, the booking date, and the total price.

```
mysql> create view v_customer_bookings_summary as
    -> select
    ->     concat(c.first_name, ' ', c.last_name) as customer_full_name,
    ->     p.package_name,
    ->     b.booking_date,
    ->     b.total_price
    -> from bookings b
    -> join customers c on b.customer_id = c.customer_id
    -> join packages p on b.package_id = p.package_id;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> select * from v_customer_bookings_summary;
+--------------------+-------------------+--------------+-------------+
| customer_full_name | package_name      | booking_date | total_price |
+--------------------+-------------------+--------------+-------------+
| Amit Sharma        | Europe Tour       | 2026-01-03   |    60000.00 |
| Sneha Patil        | Dubai Trip        | 2026-02-03   |    45000.00 |
| Rahul Verma        | Europe Tour       | 2025-11-03   |    70000.00 |
| Priya Mehta        | Thailand Escape   | 2025-08-03   |    30000.00 |
| Rohan Kulkarni     | Singapore Special | 2025-12-03   |    55000.00 |
+--------------------+-------------------+--------------+-------------+
5 rows in set (0.00 sec)
```

6. Data Integrity & Control (TCL Commands)

The IT team is performing a critical update on booking records. They need to update a few booking prices based on a new pricing model. The team must handle this process within a transaction to guarantee data consistency.

- **Start a new transaction.**
- **Update the total_price for two specific bookings (e.g., change booking_id 701 to $1500 and 702 to $2000).**
- **After the first price update (booking_id 701), create a savepoint called after_first_price_update.**
- **You realize you made a mistake and update booking_id 703 to $1500 (which is an incorrect price).**
- **Rollback the transaction to the after_first_price_update savepoint to undo only the erroneous change.**
- **Correctly update booking_id 703 to $2500.**

Commit the transaction to finalize all correct changes.

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> update bookings
    -> set total_price = 1500
    -> where booking_id = 701;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> savepoint after_first_price_update;
Query OK, 0 rows affected (0.00 sec)

mysql> update bookings
    -> set total_price = 2000
    -> where booking_id = 702;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update bookings
    -> set total_price = 1500
    -> where booking_id = 703;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> rollback to after_first_price_update;

Query OK, 0 rows affected (0.00 sec)


mysql> update bookings

    -> set total_price = 2500

    -> where booking_id = 703;

Query OK, 1 row affected (0.00 sec)

Rows matched: 1  Changed: 1  Warnings: 0


mysql> commit;

Query OK, 0 rows affected (0.01 sec)


mysql> select booking_id, total_price from bookings;

+------------+-------------+

| booking_id | total_price |

+------------+-------------+

|        701 |     1500.00 |

|        702 |    45000.00 |

|        703 |     2500.00 |

|        704 |    30000.00 |

|        705 |    55000.00 |

+------------+-------------+

5 rows in set (0.00 sec)
```

Command Prompt - mysql -u root -p

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\CHIMU>mysql -u root -p
Enter password: *********
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 29
Server version: 8.0.45 MySQL Community Server - GPL

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Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database voyage_ventures;
Query OK, 1 row affected (0.03 sec)

mysql> use voyage_ventures;
Database changed
mysql> create table customers (
    -> customer_id int primary key,
    ->  first_name varchar(50) not null,
    -> last_name varchar(50) not null,
    -> email varchar(100),
    -> city varchar(50)
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql> create table packages (
    ->     package_id int primary key,
    ->     package_name varchar(100),
    ->     destination varchar(100),
    ->     duration_days int
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> create table employees (
    ->     employee_id int primary key,
    ->     first_name varchar(50),
    ->     last_name varchar(50),
    ->     hire_date date,
```

Activate Windows
Go to Settings to activate Windows.

Command Prompt - mysql -u root -p

```
mysql> create table employees (
    ->     employee_id int primary key,
    ->     first_name varchar(50),
    ->     last_name varchar(50),
    ->     hire_date date,
    ->     department varchar(50)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> create table bookings (
    ->     booking_id int primary key,
    ->     customer_id int,
    ->     package_id int,
    ->     employee_id int,
    ->     booking_date date,
    ->     total_price decimal(10,2),
    ->     foreign key (customer_id) references customers(customer_id),
    ->     foreign key (package_id) references packages(package_id),
    ->     foreign key (employee_id) references employees(employee_id)
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql> desc customers;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| customer_id | int          | NO   | PRI | NULL    |       |
| first_name  | varchar(50)  | NO   |     | NULL    |       |
| last_name   | varchar(50)  | NO   |     | NULL    |       |
| email       | varchar(100) | YES  |     | NULL    |       |
| city        | varchar(50)  | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
5 rows in set (0.02 sec)

mysql> desc packages;
+---------------+--------------+------+-----+---------+-------+
| Field         | Type         | Null | Key | Default | Extra |
+---------------+--------------+------+-----+---------+-------+
| package_id    | int          | NO   | PRI | NULL    |       |
| package_name  | varchar(100) | YES  |     | NULL    |       |
| destination   | varchar(100) | YES  |     | NULL    |       |
| duration_days | int          | YES  |     | NULL    |       |
+---------------+--------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

Activate Windows
Go to Settings to activate Windows.

Command Prompt - mysql -u root -p

```
mysql> desc employees;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| employee_id | int         | NO   | PRI | NULL    |       |
| first_name  | varchar(50) | YES  |     | NULL    |       |
| last_name   | varchar(50) | YES  |     | NULL    |       |
| hire_date   | date        | YES  |     | NULL    |       |
| department  | varchar(50) | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
5 rows in set (0.01 sec)

mysql> desc bookings;
+--------------+---------------+------+-----+---------+-------+
| Field        | Type          | Null | Key | Default | Extra |
+--------------+---------------+------+-----+---------+-------+
| booking_id   | int           | NO   | PRI | NULL    |       |
| customer_id  | int           | YES  | MUL | NULL    |       |
| package_id   | int           | YES  | MUL | NULL    |       |
| employee_id  | int           | YES  | MUL | NULL    |       |
| booking_date | date          | YES  |     | NULL    |       |
| total_price  | decimal(10,2) | YES  |     | NULL    |       |
+--------------+---------------+------+-----+---------+-------+
6 rows in set (0.00 sec)

mysql> insert into customers values
    -> (101, 'Amit', 'Sharma', 'amit@gmail.com', 'Pune'),
    -> (102, 'Sneha', 'Patil', 'sneha@gmail.com', 'Mumbai'),
    -> (103, 'Rahul', 'Verma', 'rahul@gmail.com', 'Delhi'),
    -> (104, 'Priya', 'Mehta', 'priya@gmail.com', 'Chennai'),
    -> (105, 'Rohan', 'Kulkarni', 'rohan@gmail.com', 'Bangalore');
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> insert into packages values
    -> (201, 'Europe Tour', 'Paris', 12),
    -> (202, 'Dubai Trip', 'Dubai', 7),
    -> (203, 'Thailand Escape', 'Bangkok', 10),
    -> (204, 'Maldives Luxury', 'Maldives', 5),
    -> (205, 'Singapore Special', 'Singapore', 8);
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> insert into employees values
```

Command Prompt - mysql -u root -p

```
mysql> insert into employees values
    -> (301, 'Neha', 'Joshi', '2023-01-15', 'Sales'),
    -> (302, 'Arjun', 'Reddy', '2022-03-10', 'Marketing'),
    -> (303, 'Kiran', 'Desai', '2021-07-20', 'Sales'),
    -> (304, 'Meera', 'Kapoor', '2024-02-01', 'Support');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> insert into bookings values
    -> (701, 101, 201, 301, curdate() - interval 2 month, 60000),
    -> (702, 102, 202, 302, curdate() - interval 1 month, 45000),
    -> (703, 103, 201, 301, curdate() - interval 4 month, 70000),
    -> (704, 104, 203, 303, curdate() - interval 7 month, 30000),
    -> (705, 105, 205, 304, curdate() - interval 3 month, 55000);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from customers;
+-------------+------------+-----------+-----------------+-----------+
| customer_id | first_name | last_name | email           | city      |
+-------------+------------+-----------+-----------------+-----------+
|         101 | Amit       | Sharma    | amit@gmail.com  | Pune      |
|         102 | Sneha      | Patil     | sneha@gmail.com | Mumbai    |
|         103 | Rahul      | Verma     | rahul@gmail.com | Delhi     |
|         104 | Priya      | Mehta     | priya@gmail.com | Chennai   |
|         105 | Rohan      | Kulkarni  | rohan@gmail.com | Bangalore |
+-------------+------------+-----------+-----------------+-----------+
5 rows in set (0.00 sec)

mysql> select * from packages;
+------------+-------------------+-------------+---------------+
| package_id | package_name      | destination | duration_days |
+------------+-------------------+-------------+---------------+
|        201 | Europe Tour       | Paris       |            12 |
|        202 | Dubai Trip        | Dubai       |             7 |
|        203 | Thailand Escape   | Bangkok     |            10 |
|        204 | Maldives Luxury   | Maldives    |             5 |
|        205 | Singapore Special | Singapore   |             8 |
+------------+-------------------+-------------+---------------+
5 rows in set (0.00 sec)

mysql> select * from employees;
```

Command Prompt - mysql -u root -p

```
mysql> select * from employees;
+-------------+------------+-----------+------------+------------+
| employee_id | first_name | last_name | hire_date  | department |
+-------------+------------+-----------+------------+------------+
|         301 | Neha       | Joshi     | 2023-01-15 | Sales      |
|         302 | Arjun      | Reddy     | 2022-03-10 | Marketing  |
|         303 | Kiran      | Desai     | 2021-07-20 | Sales      |
|         304 | Meera      | Kapoor    | 2024-02-01 | Support    |
+-------------+------------+-----------+------------+------------+
4 rows in set (0.00 sec)

mysql> select * from bookings;
+------------+-------------+------------+-------------+--------------+-------------+
| booking_id | customer_id | package_id | employee_id | booking_date | total_price |
+------------+-------------+------------+-------------+--------------+-------------+
|        701 |         101 |        201 |         301 | 2026-01-03   |    60000.00 |
|        702 |         102 |        202 |         302 | 2026-02-03   |    45000.00 |
|        703 |         103 |        201 |         301 | 2025-11-03   |    70000.00 |
|        704 |         104 |        203 |         303 | 2025-08-03   |    30000.00 |
|        705 |         105 |        205 |         304 | 2025-12-03   |    55000.00 |
+------------+-------------+------------+-------------+--------------+-------------+
5 rows in set (0.00 sec)

mysql> select
    ->     lower(concat(c.first_name, ' ', c.last_name)) as customer_full_name,
    ->     concat('Destination: ', p.destination) as destination,
    ->     date_add(b.booking_date, interval p.duration_days day) as estimated_return_date
    -> from bookings b
    -> join customers c on b.customer_id = c.customer_id
    -> join packages p on b.package_id = p.package_id
    -> where b.booking_date >= date_sub(curdate(), interval 6 month)
    -> order by b.booking_date desc;
+--------------------+------------------------+-----------------------+
| customer_full_name | destination            | estimated_return_date |
+--------------------+------------------------+-----------------------+
| sneha patil        | Destination: Dubai     | 2026-02-10            |
| amit sharma        | Destination: Paris     | 2026-01-15            |
| rohan kulkarni     | Destination: Singapore | 2025-12-11            |
| rahul verma        | Destination: Paris     | 2025-11-15            |
+--------------------+------------------------+-----------------------+
4 rows in set (0.01 sec)

mysql> select
    ->     p.destination,
```


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Command Prompt - mysql -u root -p

```
mysql> select
    ->     p.destination,
    ->     sum(b.total_price) as total_revenue,
    ->     avg(b.total_price) as average_booking_price,
    ->     count(b.booking_id) as total_bookings
    -> from bookings b
    -> join packages p on b.package_id = p.package_id
    -> group by p.destination
    -> having sum(b.total_price) > 50000;
+-------------+---------------+-----------------------+----------------+
| destination | total_revenue | average_booking_price | total_bookings |
+-------------+---------------+-----------------------+----------------+
| Paris       |     130000.00 |          65000.000000 |              2 |
| Singapore   |      55000.00 |          55000.000000 |              1 |
+-------------+---------------+-----------------------+----------------+
2 rows in set (0.01 sec)

mysql> select distinct
    ->     e.first_name,
    ->     e.last_name,
    ->     e.department
    -> from employees e
    -> join bookings b on e.employee_id = b.employee_id
    -> where b.customer_id in (
    ->     select customer_id
    ->     from bookings
    ->     group by customer_id
    ->     having sum(total_price) > (
    ->         select avg(total_spend)
    ->         from (
    ->             select sum(total_price) as total_spend
    ->             from bookings
    ->             group by customer_id
    ->         ) as temp
    ->     )
    -> );
+------------+-----------+------------+
| first_name | last_name | department |
+------------+-----------+------------+
| Neha       | Joshi     | Sales      |
| Meera      | Kapoor    | Support    |
+------------+-----------+------------+
2 rows in set (0.01 sec)
```


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Command Prompt - mysql -u root -p

```
2 rows in set (0.01 sec)

mysql> select first_name, last_name
    -> from customers
    -> where customer_id in (
    ->     select b.customer_id
    ->     from bookings b
    ->     join packages p on b.package_id = p.package_id
    ->     where p.duration_days > 10
    -> )
    ->
    -> union
    ->
    -> select first_name, last_name
    -> from employees
    -> where hire_date >= date_sub(curdate(), interval 2 year);
+------------+-----------+
| first_name | last_name |
+------------+-----------+
| Amit       | Sharma    |
| Rahul      | Verma     |
+------------+-----------+
2 rows in set (0.00 sec)

mysql> create view v_customer_bookings_summary as
    -> select
    ->     concat(c.first_name, ' ', c.last_name) as customer_full_name,
    ->     p.package_name,
    ->     b.booking_date,
    ->     b.total_price
    -> from bookings b
    -> join customers c on b.customer_id = c.customer_id
    -> join packages p on b.package_id = p.package_id;
Query OK, 0 rows affected (0.02 sec)

mysql> select * from v_customer_bookings_summary;
+--------------------+-------------------+--------------+-------------+
| customer_full_name | package_name      | booking_date | total_price |
+--------------------+-------------------+--------------+-------------+
| Amit Sharma        | Europe Tour       | 2026-01-03   |    60000.00 |
| Sneha Patil        | Dubai Trip        | 2026-02-03   |    45000.00 |
| Rahul Verma        | Europe Tour       | 2025-11-03   |    70000.00 |
| Priya Mehta        | Thailand Escape   | 2025-08-03   |    30000.00 |
| Rohan Kulkarni     | Singapore Special | 2025-12-03   |    55000.00 |
```

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Command Prompt - mysql -u root -p

```
Query OK, 0 rows affected (0.02 sec)

mysql> select * from v_customer_bookings_summary;
+--------------------+-------------------+--------------+-------------+
| customer_full_name | package_name      | booking_date | total_price |
+--------------------+-------------------+--------------+-------------+
| Amit Sharma        | Europe Tour       | 2026-01-03   |    60000.00 |
| Sneha Patil        | Dubai Trip        | 2026-02-03   |    45000.00 |
| Rahul Verma        | Europe Tour       | 2025-11-03   |    70000.00 |
| Priya Mehta        | Thailand Escape   | 2025-08-03   |    30000.00 |
| Rohan Kulkarni     | Singapore Special | 2025-12-03   |    55000.00 |
+--------------------+-------------------+--------------+-------------+
5 rows in set (0.00 sec)

mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> update bookings
    -> set total_price = 1500where booking_id = 701;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'booking_id =
701' at line 2
mysql> update bookings
    -> set total_price = 1500
    -> where booking_id = 701;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> savepoint after_first_price_update;
Query OK, 0 rows affected (0.00 sec)

mysql> update bookings
    -> set total_price = 2000
    -> where booking_id = 702;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update bookings
    -> set total_price = 1500
    -> where booking_id = 703;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> rollback to after_first_price_update;
Query OK, 0 rows affected (0.00 sec)
```

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```
mysql> update bookings
    -> set total_price = 2000
    -> where booking_id = 702;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update bookings
    -> set total_price = 1500
    -> where booking_id = 703;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> rollback to after_first_price_update;
Query OK, 0 rows affected (0.00 sec)

mysql> update bookings
    -> set total_price = 2500
    -> where booking_id = 703;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> commit;
Query OK, 0 rows affected (0.01 sec)

mysql> select booking_id, total_price from bookings;
+------------+-------------+
| booking_id | total_price |
+------------+-------------+
|        701 |     1500.00 |
|        702 |    45000.00 |
|        703 |     2500.00 |
|        704 |    30000.00 |
|        705 |    55000.00 |
+------------+-------------+
5 rows in set (0.00 sec)

mysql> _
```

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